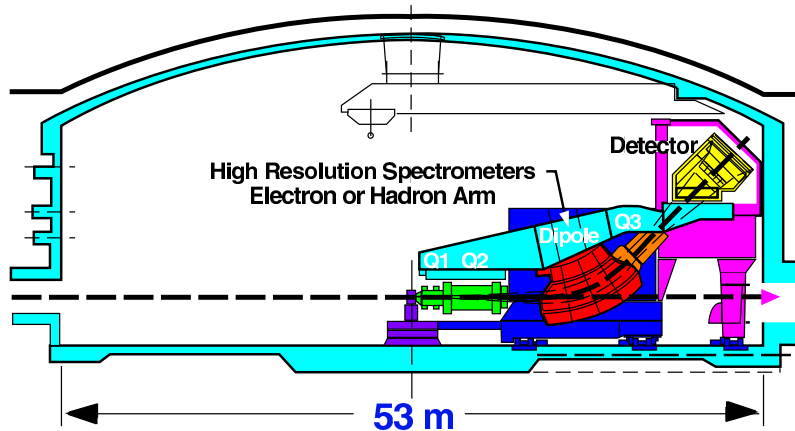


Electron Beam Polarimetry at JLab

- Experiments using polarized electrons at JLab
- JLab polarized beam
- Polarimetry at JLab:
 - Mott polarimetry
 - Møller polarimetry
 - Compton polarimetry
- Special challenges of new experiments
- Possible improvements in polarimetry (outlook)

SIMULTANEOUS COMPLEMENTARY EXPERIMENTS

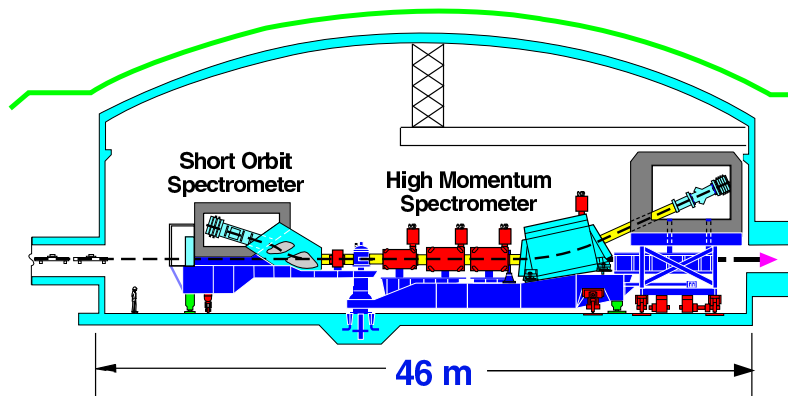
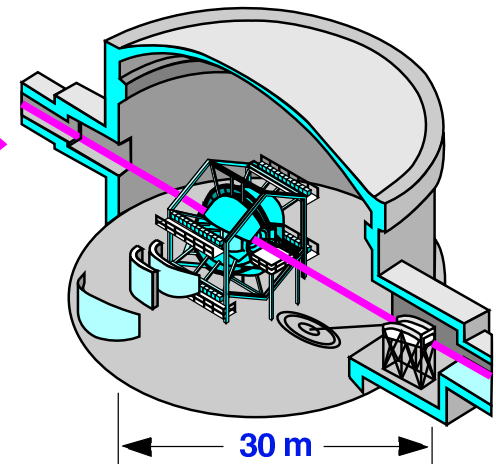


◀ HALL A

Pair of identical High Resolution Spectrometers (HRS²)

HALL B ▶

CEBAF's Large Acceptance Spectrometer (CLAS) and Bremsstrahlung Photon Tagger



◀ HALL C

High Momentum Spectrometer (HMS) and Short Orbit Spectrometer (SOS)



Polarization Experiments at JLab

More than half of all experiments in Halls A/B/C are using the longitudinal beam polarization:

Single Spin: Parity Violation Experiments

- Parity violation
- $\vec{e}^- p, A \rightarrow e^- p, A$ elastic: formafactors HAPPEX, G0, SM QWEAK
- $\vec{e}^- A \rightarrow e^- A$ elastic: nuclear physics ^{208}Pb
- $\vec{e}^- p \rightarrow e^- X$ inelastic: SM

$$A_{obs} \propto \mathcal{P}_{beam} \times A_{reaction}$$

Double Spin Experiments

- $\vec{e}^- p \rightarrow e^- \vec{p}$ elastic: formafactors G_E^p
- $\vec{e}^- \vec{N} \rightarrow e^- N$ elastic: formafactors G_E^n
- $\vec{e}^- \vec{N} \rightarrow e^- X$ inelastic: structure functions, GDH

$$A_{obs} \propto \mathcal{P}_{beam} \times \mathcal{P}_{target} \times A_{reaction}$$

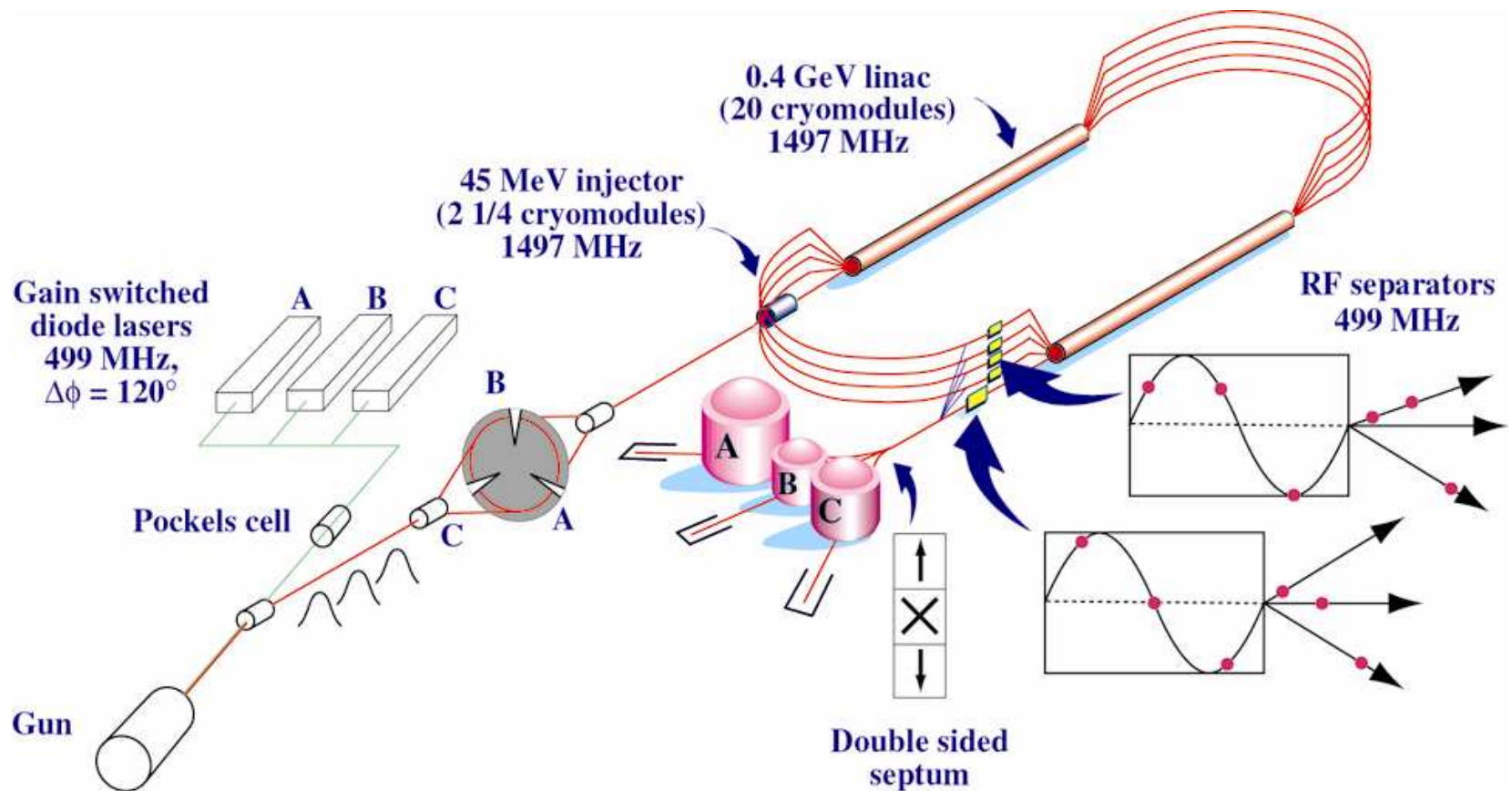
$$A_{obs} \propto \mathcal{P}_{beam} \times A_{recoil} \times A_{reaction}$$

Typically, $\sigma \mathcal{P}_{beam} / \mathcal{P}_{beam} < \sigma \mathcal{P}_{target} / \mathcal{P}_{target}$

Features of CEBAF Operations

- Beam energy 0.8 - 6.0 GeV, current 0.01 - 100 μ A
- 3 Halls are running simultaneously with different beam energies and currents
 - The beam may interfere with beams to other halls
 - Spin precession in different halls is different
- Operations, invasive for other halls are better avoided

Continuous Electron Beam Accelerator Facility

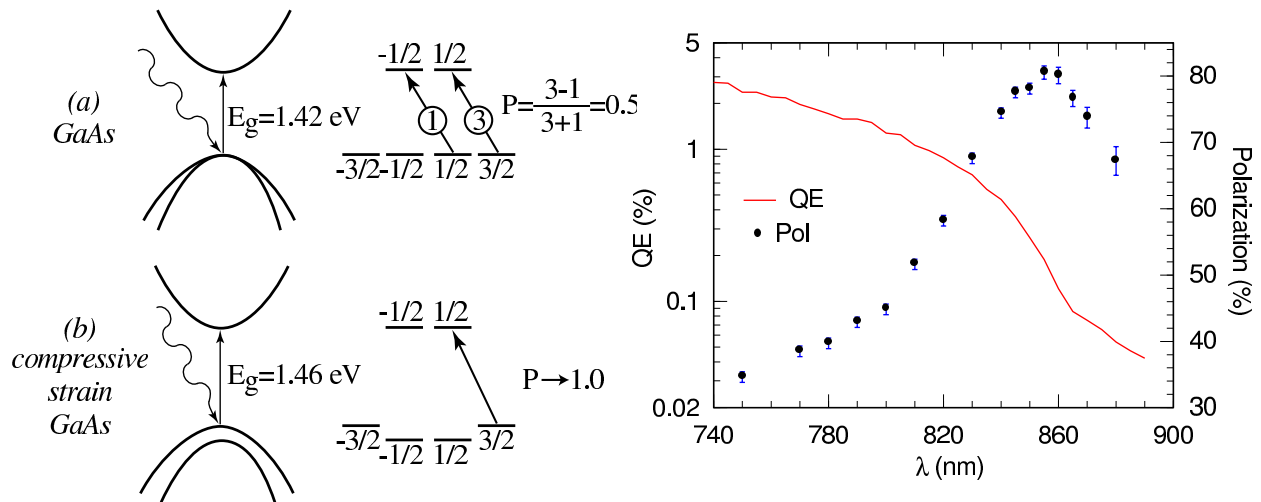


Thomas Jefferson National Accelerator Facility

Operated by the Southeastern Universities Research Association for the U.S. Depart. Of Energy

OpsReviewCWL:mbs Ops Review, Jan. 22-25, 2002,
1

Features of CEBAF Injector



- Strained GaAs provides $\mathcal{P} \sim 80\%$, $I < 80 \mu\text{A}$
- 30Hz helicity flip
- 2 ways to attenuate the beam:
 - Laser light attenuation
 - Electron chopper slit
- Opposite beam polarization in some halls
- Laser dark current $\Rightarrow$ leak of a wrong laser through the slit

The beam polarization may be diluted by a leak through from another hall beam.

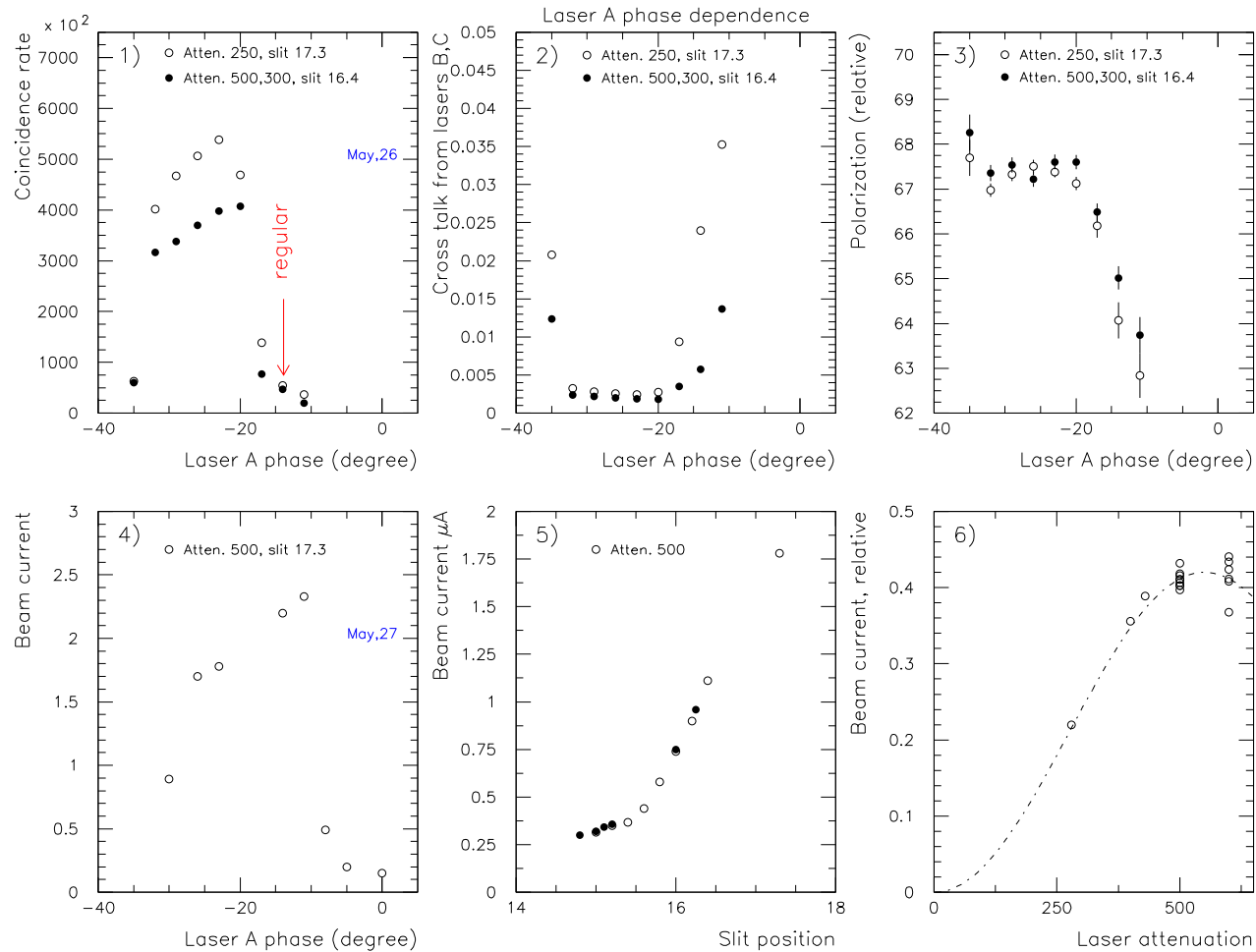
- Current dependent
- Depends on the other hall operation

The beam polarization may depend on the current

Electron Beam Polarimetry at JLab

CEBAF Polarization dependence on laser phase

99/07/06 18.53



Electron Polarimetry at JLab

Injector

- Mott polarimeter at 5 MeV, $\sigma_{syst} \sim 3\% \Rightarrow 1\%$ (?)

Hall A

- Møller polarimeter, 2 arm 0.8-6.0 GeV, $\sigma_{syst} \sim 3\%$
- Compton polarimeter > 2 GeV, $\sigma_{syst} \sim 1-2\%$

Hall B

- Møller polarimeter, 2 arm 0.8-6.0 GeV $\sigma_{syst} \sim ?$

Hall C

- Møller polarimeter, 2 arm 0.8-6.0 GeV
 $\sigma_{syst} \sim 1.5(0.5)\%$

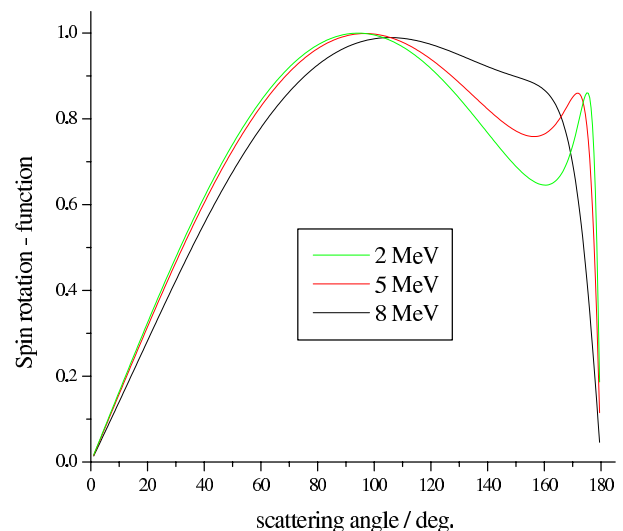
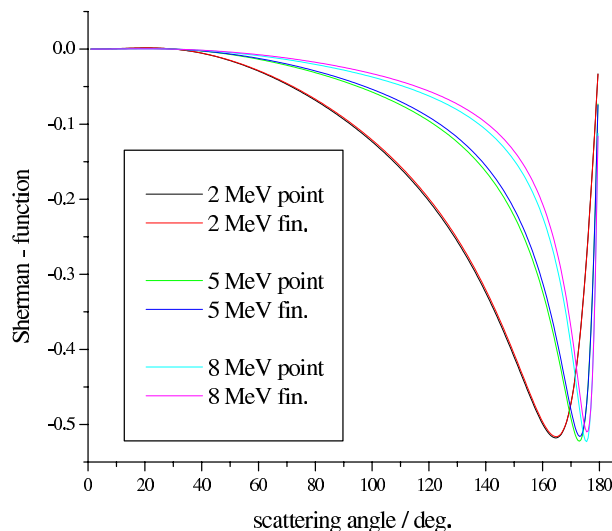
- Description of the polarimeters
- Cross-calibration of the polarimeters

Mott Scattering

0.1-10 MeV: $e^- \uparrow + Au \rightarrow e^- + Au$

Analyzing power - Sherman functions $\sim 1-3\%$:

- Dirac equation on a pure Coulomb potential
- Nucleus thickness: phase shifts of scat. amplitudes
- Spin rotation functions
- Other: electron screening, radiative corrections etc

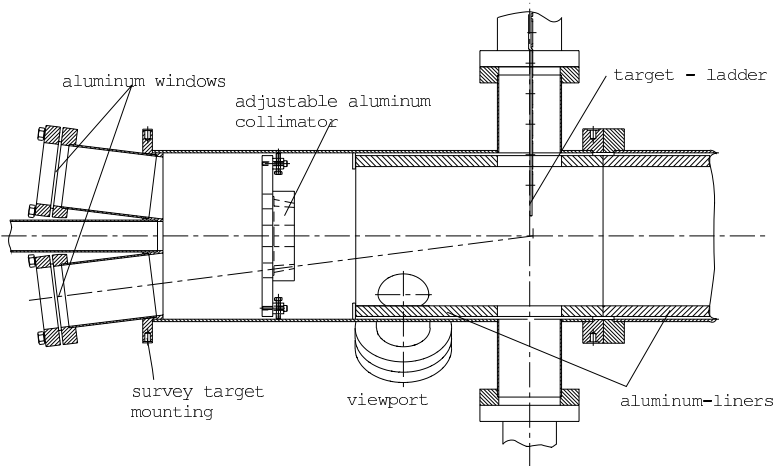


- Multiple and plural scattering
- No energy loss should be allowed
- Single arm - backgrounds are important

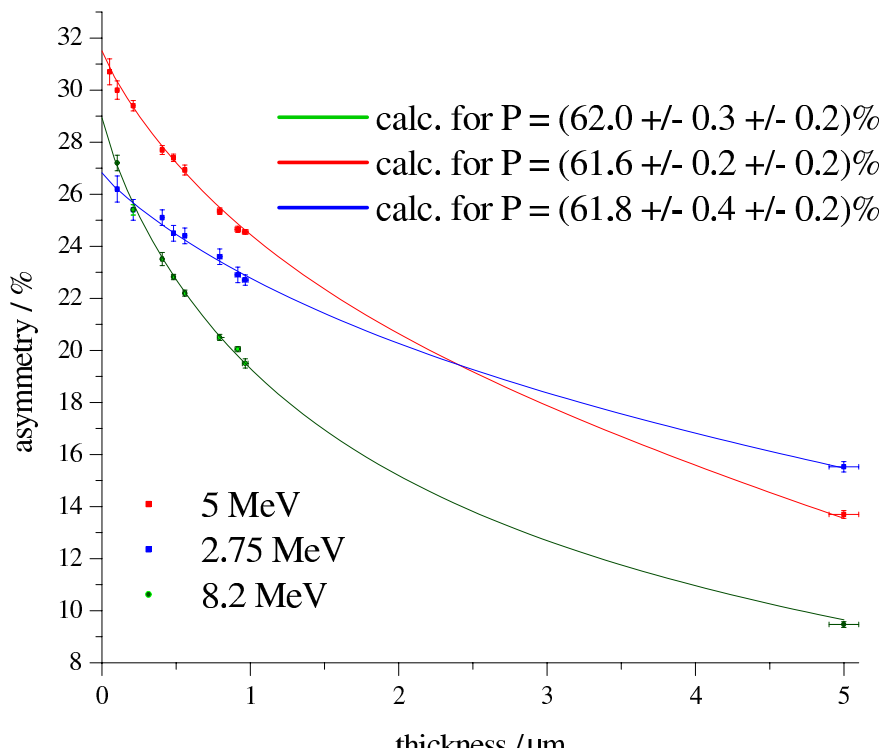
Typical target: Au foils $0.01-1.00\mu\text{m}$ thick $\Rightarrow$ 0 thickness

Mott Polarimeter at JLab

Measurements 5 MeV and $\sim 1 \mu\text{A}$



Careful shielding
Veto detectors
Energy measured



$$\sigma(P)/P = 1\%(Sherman) \oplus 0.5\%(other) \text{ (unpublished)}$$

- Low current measurement
- Invasive for all 3 halls at CEBAF

Møller Scattering

$\vec{e}^- + \vec{e}^- \rightarrow e^- + e^-$ QED (unpolarized: Møller, 1932).

$$\frac{d\sigma}{d\Omega_{CM}} = \frac{d\sigma_0}{d\Omega_{CM}} \cdot \left(1 + \sum_{i=X,Y,Z} (A_{ii} \cdot \mathcal{P}_{target\ i} \cdot \mathcal{P}_{beam\ i})\right)$$

$$\frac{d\sigma_0}{d\Omega_{CM}} \approx \frac{r_e^2}{4\gamma^2} \cdot \frac{(4 - \sin^2 \theta_{CM})^2}{\sin^4 \theta_{CM}} \text{ in CM, or}$$

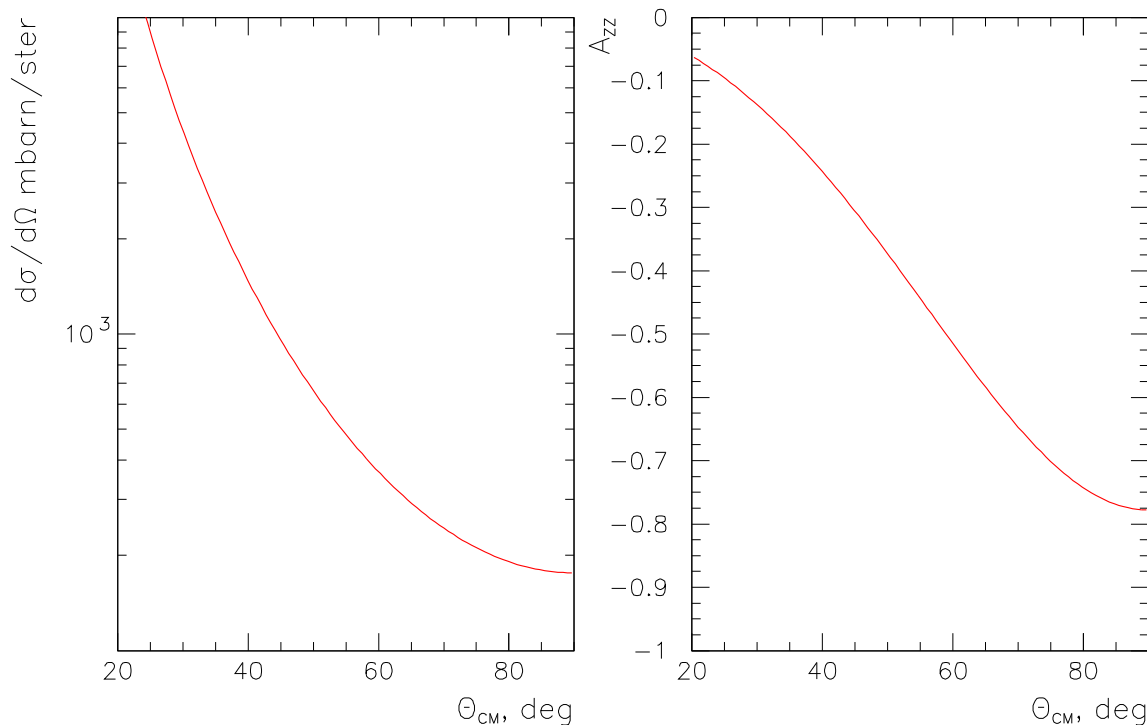
$$\frac{d\sigma_0}{d\Omega} \big|_{\theta_{CM}=90^\circ} \approx 176 \text{ mbarn/ster in LAB}$$

Asymmetry:

$$A_{ZZ} = -\frac{\sin^2 \theta_{CM} \cdot (7 + \cos^2 \theta_{CM})}{(3 + \cos^2 \theta_{CM})^2}$$

$$A_{XX} = -\frac{\sin^4 \theta_{CM}}{(3 + \cos^2 \theta_{CM})^2}, \quad A_{YY} = -A_{XX}$$

$$\text{At } \theta_{CM} = 90^\circ \quad A_{ZZ} = 7/9, \quad A_{XX} = -A_{YY} = -1/9$$



Møller Polarimetry

Advantages:

- High analyzing power at $\theta_{CM} = 90^\circ$ $A_{ZZ} = 7/9$
- $\frac{dA_{ZZ}}{d\theta_{CM}} \big|_{\theta_{CM}=90^\circ} = 0$ - small systematics
- Large cross-section
- 2 particles in the final state with $E \sim E_0/2$:
coincidence eliminates backgrounds

Disadvantages mainly come from the target choice of magnetized ferromagnetics :

- Relatively low polarization $\sim 8\%$
- Beam current limit ($\sim 1 \mu\text{A}$) due to target heating
- Systematic errors on the target polarization
- Kinematic distortion of scattering on K,L-shell electrons (“Levchuk effect”)

Ferromagnetic Targets

So far, Møller polarimeters used ferromagnetic foils, magnetized in an external field, for the target.

Features of the Fe group:

- 3D shell only partially filled: $3d^6$ - 4 electrons missing
- Overlap of the wave-functions with neighbour atoms and electron gas:
"exchange force" aligning spins on $3d^6$ in the SAME direction. In Fe atom, $2.22 e^-$ aligned at saturation
- "Quenching" - orbital momentum contribution cancelled,
magnetization is caused by the spins
- Magnetization drops with temperature and comes to 0 at $T_c \sim 800^\circ C$

Materials used:

- Pure Fe - best understood and studied
- Alloys, like permendur: Fe(49%), Co(49%), V(2%) - more easily magnetized

Ferromagnetic Targets: Polarization

$$\mathcal{P}_{foil} = B_{foil} \cdot \frac{g' - 1}{2\pi g'} \cdot \frac{1}{N_e \mu_B}, \text{ where}$$

B_{foil} - magnetic field in the foil

$g' = 1.900 \pm 0.005$ - spin/orbital correction, supermendur

$g' = 1.919 \pm 0.002$ - spin/orbital correction, Fe

$N_e = \rho \cdot Av \cdot Z/A$ - electron density

μ_B - Bohr magneton

Magnetization in an external field

Boundary conditions on a surface:

$$B_{perpend} = const \text{ and } H_{parallel} = const$$

Two ways to magnetized the target foil are used:

- $B_{parall} \sim 100 - 300 \text{ Gs}$, foil at $\sim 20^\circ$
- $B_{perp} \sim 3 - 4 \text{ T}$ “brute force”

Foils in “Weak” Field

- Used everywhere, except in Hall C at JLab
- External $B \sim 80 - 600 \text{ Gs}$, parallel to the beam
- Target foil $10\text{-}100 \mu\text{m}$ thick at $\sim 20^\circ$ to the beam
- Important: annealing, mechanical treatment

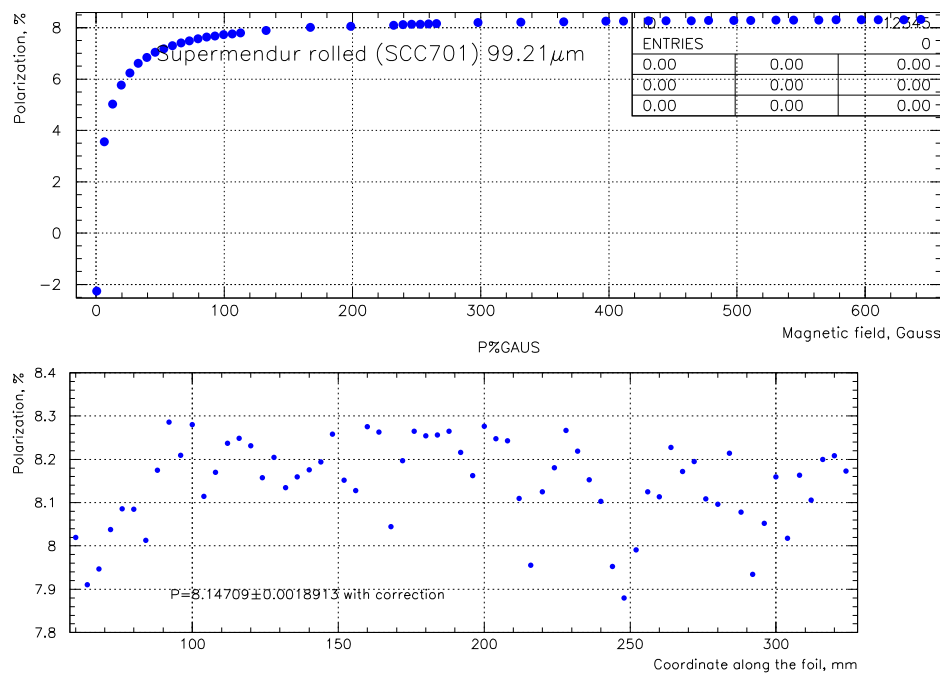
The magnetic flux through the foil can be measured with a pickup coil wound around the foil, and:

$$\Phi = \int \varepsilon(t) dt, \quad B_{\text{foil}} = \Phi / (\text{width} \cdot \text{thick}) = \Phi \cdot \rho \cdot \text{length} / \text{weight}$$

Measuring Φ : different methods.

Hall A: constant field, removing the foil from the coil

$$\mathcal{P}_{\text{foil}} = \frac{g' - 1}{2\pi g'} \cdot \frac{1}{\mu_B} \cdot \Phi_{\text{foil}} \cdot \frac{\text{length} \cdot A}{\text{weight} \cdot Av \cdot Z}$$



- Measured: an average over a large surface
- Foil's non-uniformity
- At the end, different foils compared give $\sigma(\mathcal{P})/\mathcal{P} \sim 3\%$

Foils in Saturating Fields

- Used in Hall C at JLab
- External $B \sim 4\text{ T}$, parallel to the beam
- Target foils $4\text{-}10\text{ }\mu\text{m}$ thick perpendicular to the beam
- Important: annealing, mechanical treatment

The foil should be fully saturated.

Attempts to measure the magnetization with the Kerr effect: rotation of the polarization plane of the reflected light.

Problems:

- Rotation angle is a fraction of a degree only
- Field on the surface and in the bulk may be different (band theory)

Good relative measurements have been done: saturation at 2.8 Tm , temperature dependence.

The magnetization value was taken from literature.

Claim: $\sigma(\mathcal{P})/\mathcal{P} \sim 0.25\%$

Møller Polarimeter Optics

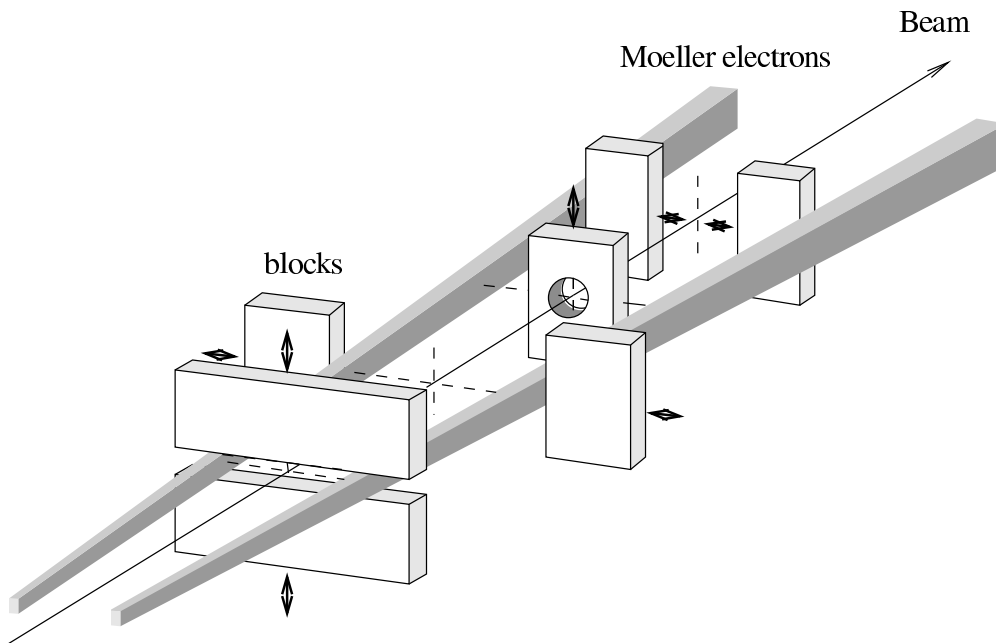
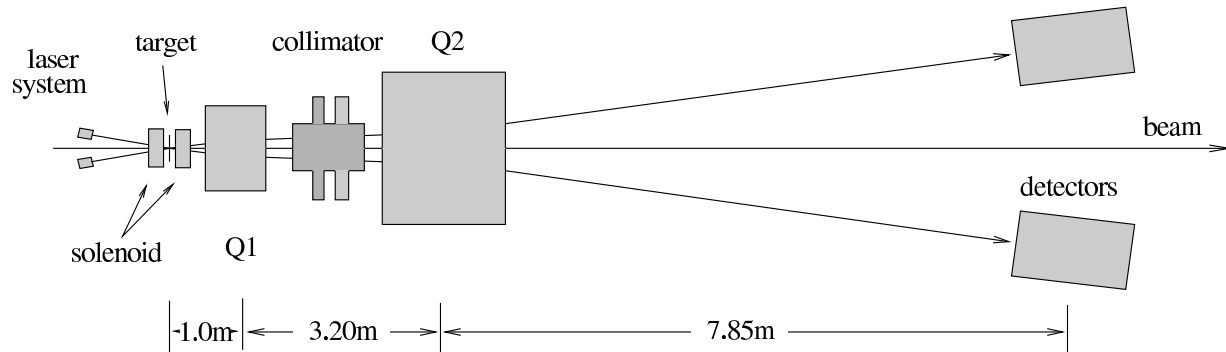
- Single arm used with pulsed beams (SLAC)
BG~10-30% from radiative Mott
- Double arm - practically no background

Optics selection:

- Use Møller θ -p correlation: quad focussing
- Use Møller $E \sim E_0/2$: dipole deflection
- Select an acceptance of $80^\circ < \theta_{CM} < 100^\circ$ or so

Møller Polarimeter Optics: JLab Hall C

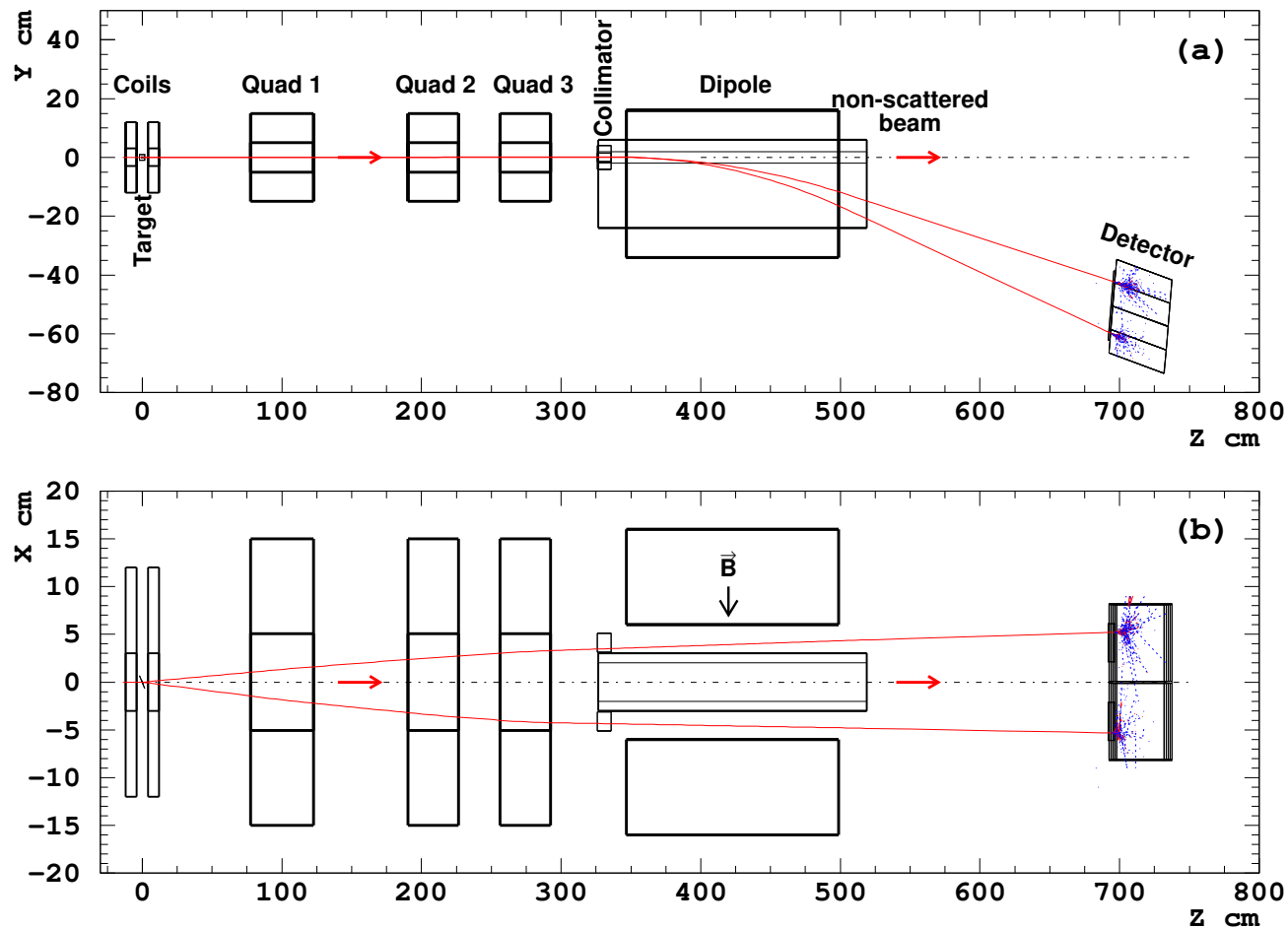
2 quadrupole magnets and many movable collimators.



Electron Beam Polarimetry at JLab

Møller Polarimeter Optics: JLab Hall A

2-3 quadrupole magnets and a dipole.



Levchuk Effect

Noticed in 1994. Many polarimetry measurements done before were wrong by 5-10%!

Main points:

- Scattering on K-shells distorts the θ -p correlation
- A strong optical focussing in exploiting the θ -p correlation
- Acceptance of electrons, scattered on K-shells (unpolarized) and D-shells (polarized) is different ($K < D$):

the effective target polarization is larger!

A correction is needed, or the optics should minimize the effect (Hall A).

Møller Hall A: Systematic Errors

$$A_{\text{obser}} = \frac{N_{+-} - N_{-+}}{N_{++} + N_{--}} \approx \overline{A_{ZZ}} \mathcal{P}_{\text{targ}} \mathcal{P}_{\text{beam}} \cdot \kappa_{\text{transv}} \kappa_{BG} \kappa_{DT} \kappa_{Lev} + \Delta_{\text{false}}$$

$\kappa = (1 + \Delta) \approx 1$ - corrections

Statistical error: $\pm 1\%$ per ~ 2 min

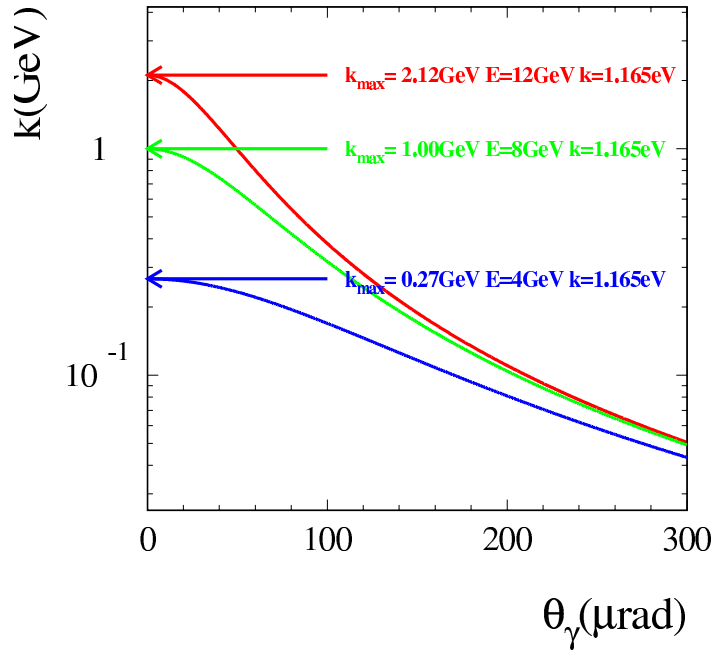
- $\overline{A_{ZZ}} \sim 0.758 - 0.772$ - simulation $\pm 0.3\%$
 - $\mathcal{P}_{\text{targ}} = \mathcal{P}_{\text{foil}} \cdot \cos \alpha_{\text{targ}}$ - target polarization
 - $\mathcal{P}_{\text{foil}}$ - magnetization measurement $\pm 3\%$
 - $\cos \alpha_{\text{targ}}$ target-beam angle $\sim 20^\circ$ & 160° $\pm 0.5\%$
 - Δ_{transv} influence of transverse polarization,
Measuring at $\alpha_{\text{targ}} \sim 25^\circ$ & 155° helps $\pm 0.2\%$
 - Δ_{BG} non-polarized background, $\sim 30 - 70\%$ in single
arm, depending on E_{beam} , low in coincidence.
Accidental coincidences are subtracted. $< 1\%$
 - Δ_{DT} dead time of electronics $< 0.3\%$
 - Levchuk effect 0.3%
 - Observed fluctuations, (beam current?) 1%
-
- Total systematic error: 3.4%

Møller Hall C: Systematic Errors

source	uncertainty	A error
beam position X	0.5 mm	0.15%
beam position Y	0.5 mm	0.03%
beam direction X	0.15 mr	0.04%
beam direction Y	0.15 mr	0.04%
current Q1	2%	0.10%
current Q2	1%	0.07%
position Q2	1 mm	0.02%
multiple scattering	10 %	0.12%
Levchuk effect	10 %	0.30%
position collimator	0.5 mm	0.06%
target temperature	50 %	0.05%
direction of B-field	2°	0.06%
value of B-field	5%	0.03%
target polarization		0.25%
dead time		?
beam current		?
total		0.25%

Compton Polarimetry

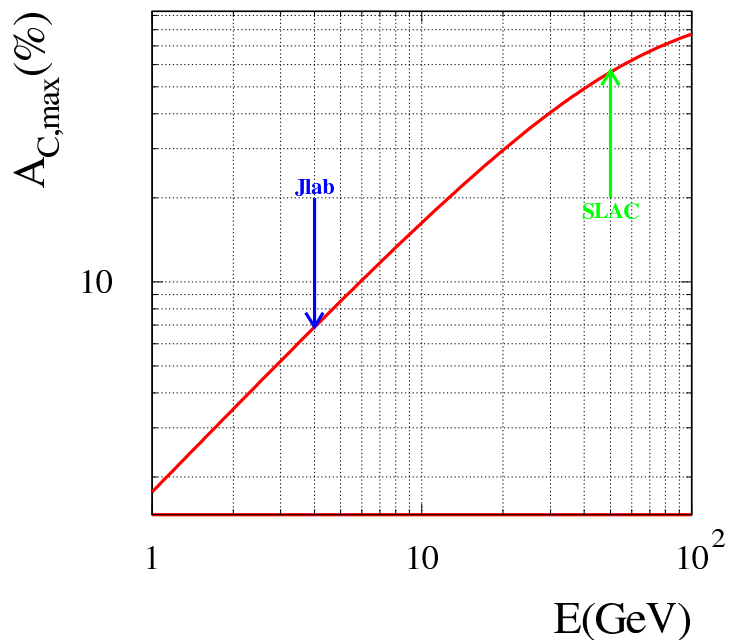
$$\vec{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma \text{ QED.}$$



Compton edge:

$$k'_{max} \approx 4y\gamma^2 k,$$

$$y = \frac{1}{1+4k\gamma/m} \sim 1$$



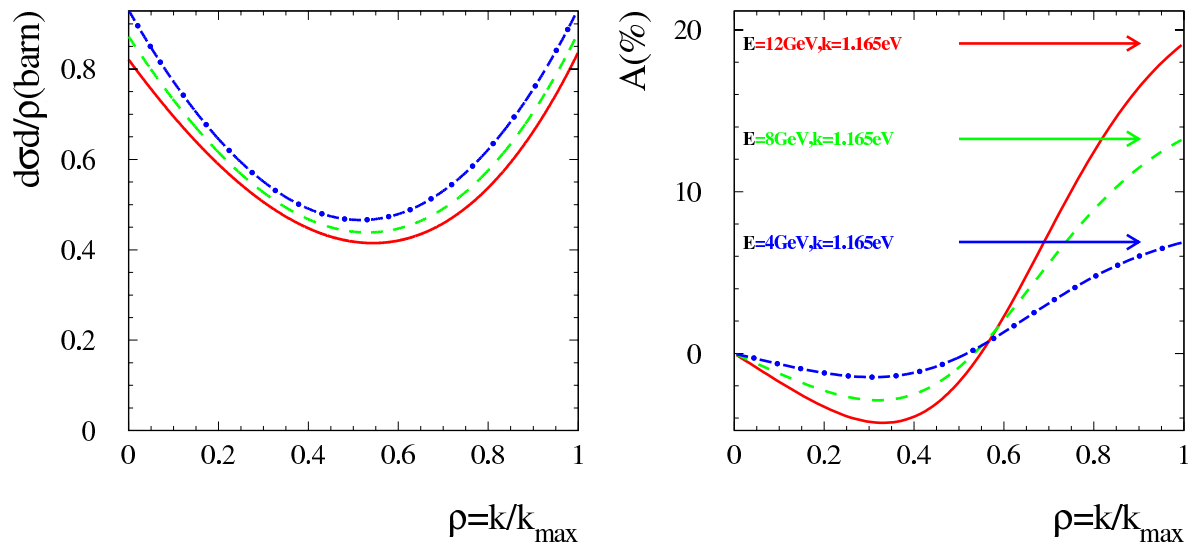
Asymmetry:

$$A \propto kE,$$

Max asymmetry:

$$A_{max} \approx \frac{1-y^2}{1+y^2},$$

Compton Polarimetry



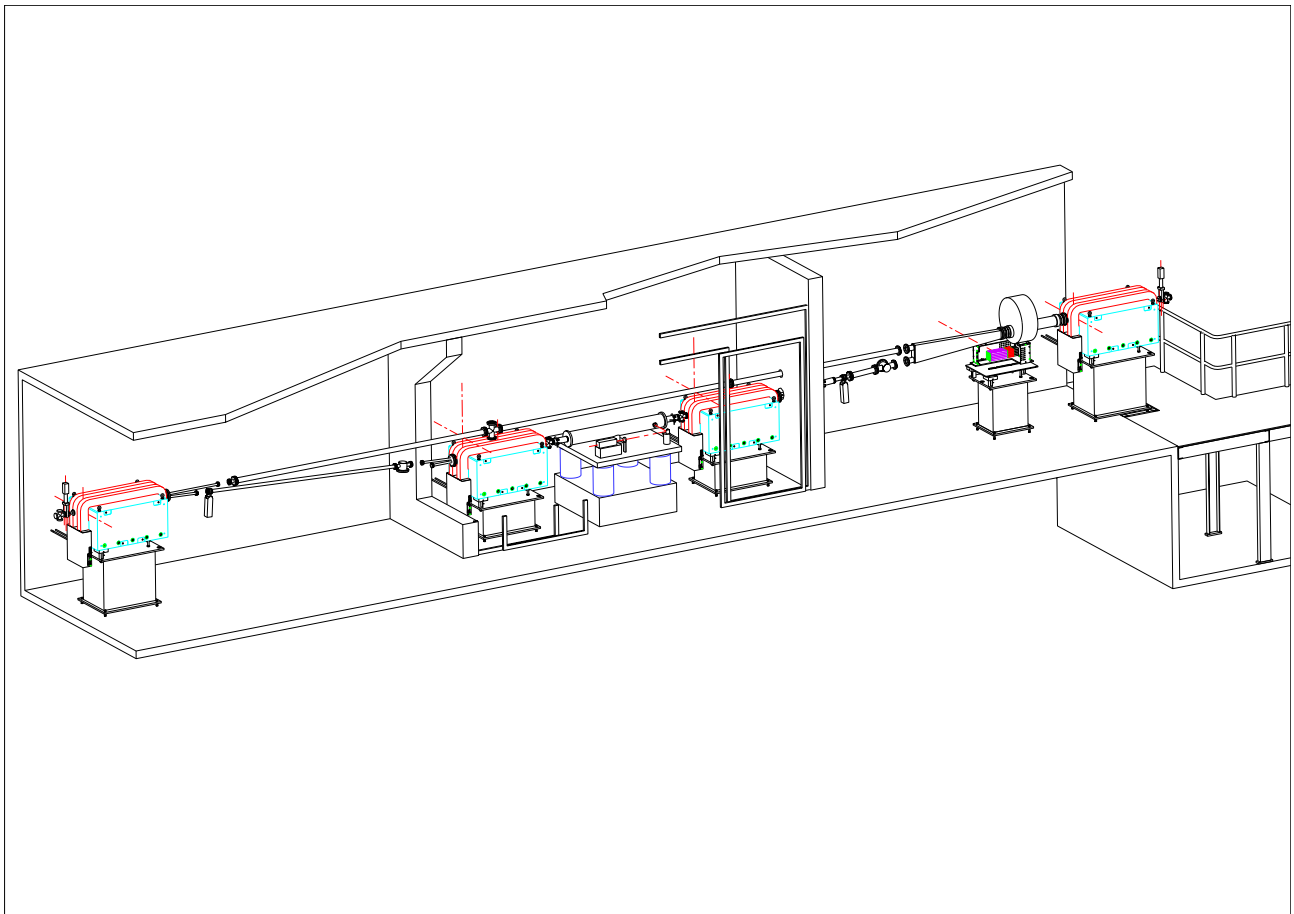
- Detecting the γ at 0 angle
- Detecting the e^- with an energy loss
- Strong $\frac{dA}{dk'}$ - good energy resolution for photons
- Photon energy cutoff
- Time needed for a measurement:

$$T \propto 1/(\sigma \cdot A^2) \propto 1/k^3 \times 1/E^4$$
- Small crossing angle needed
- Non-invasive measurement

Very good polarimetry at high energy or/and high current (storage rings)

Compton Polarimeter in Hall A

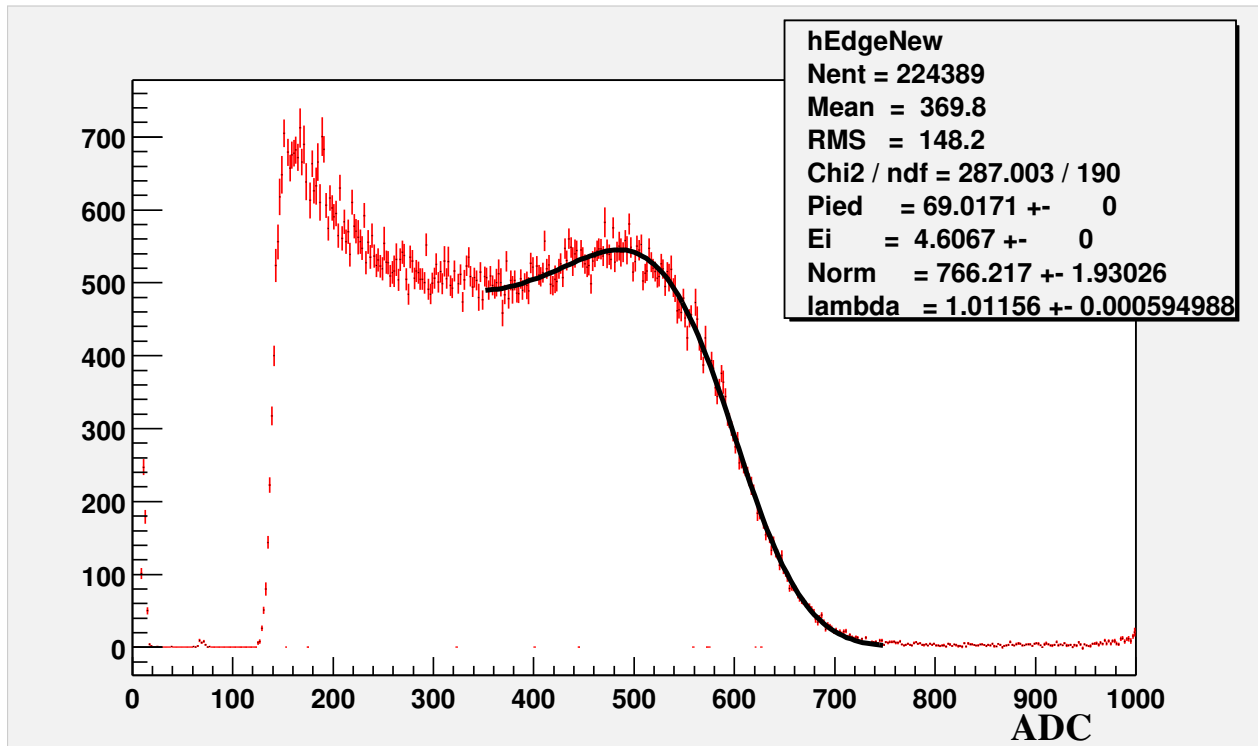
- Laser Nd:YAG 230 mW, $E_{h\nu}=1.165$ eV, $E\lambda=1064$ nm
- Monolithic 85 cm Fabry-Pérot cavity: $G\sim 8000$, 1500 W, $\mathcal{P}\sim 99.3\%$
- Crossing angle 23 mrad
- Chicane of 4 dipoles for the beam
- Photon detector at 0° $2 \times 2 \times 23$ cm³ PbWO 5×5
- Electron detector Si μ -strip 48×600 $\mu\text{m} \times 4$ planes



Photon Detector Resolution

The electron detector is used for calibration. Limitations:

- μ -strip pitch $\Rightarrow \sigma_E \sim 5 \text{ MeV}$
- Low energy - acceptance loss

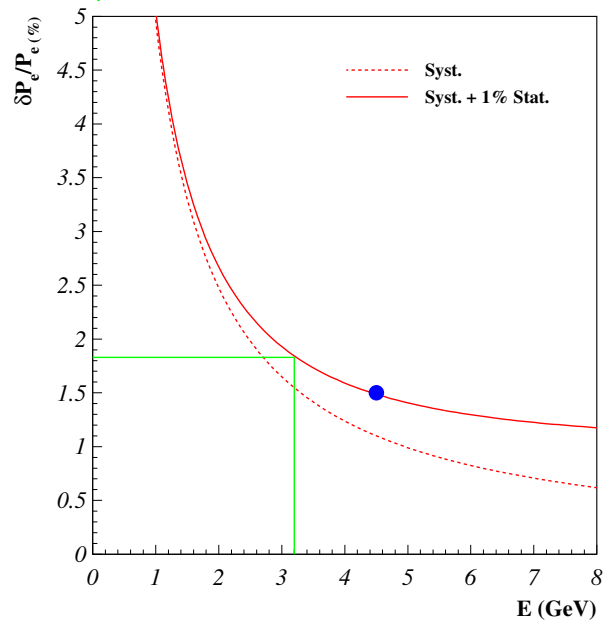


- Good agreement with the observed spectrum
- Gain drift correction

Compton Errors (Hall A)

Conditions: beam 4.5 GeV, 40 μ A, 40 min run

source	A error
Asymmetry	
Statistical	0.80%
Position and angle	0.30%
Background	0.05%
Dead time	0.10%
Cuts	0.10%
Light	
Polarization	0.50%
Analyzing power	
Response function	0.45%
Calibration	0.60%
Pile up	0.45%
Rad. correction	0.26%
Total systematic	1.15%
Total	1.40%

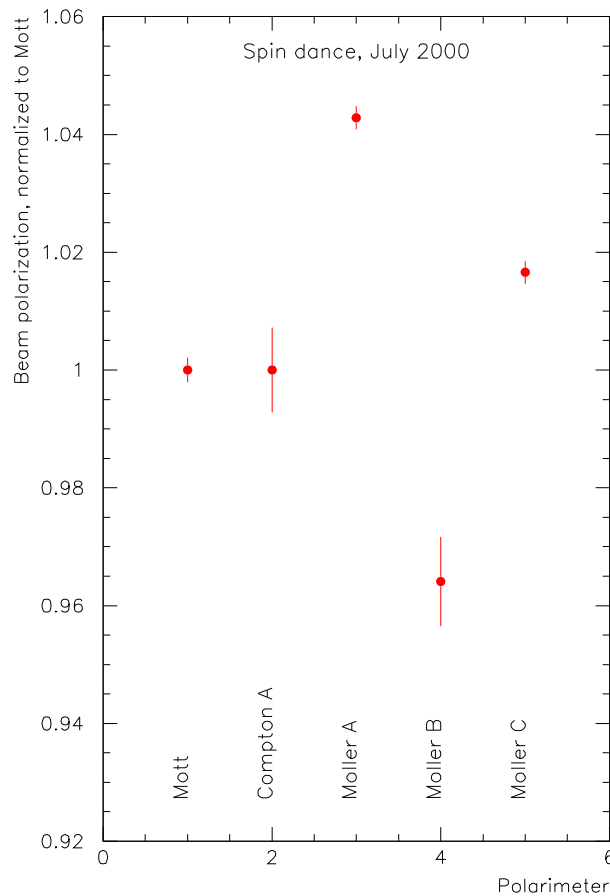


Comparison of Different Polarimetry Techniques

A comparison of polarimetry methods at 4.5 GeV.

Type	$T_{1\%}$ Stat	Syst. error	beam μA	Inva- sive?	Energy GeV
Mott	5 min	3(1)%	1	yes	0.005
Moller A	2 min	3%	0.3-1.0	yes	0.8-6
Moller C	5 min	1.5(0.5)%	0.5-1.0	yes	0.8-6
Compton	40 min	1.1%	20-100	no	>2

Cross-calibration of polarimeters



One laser: DC
Slits open
Spin rotation

Future experiments at JLab

Experiment	Syst. err without pol	Polar. error	Stat. error	Energy GeV	Comments
${}^4\text{He } \rho_s$	0.6%	2.0%	2.2%	3.2	
${}^{208}\text{Pb } n\text{-skin}$	0.5%	1.0%	3.0%	0.85	?
eP $\sin^2\theta_W$	2.4%	<1.4%	2.8%	1.16	?
DIS $\sin^2\theta_W$	0.3%	<1.0%	0.8%	10.0	distant

For 2 experiments at low energy there is no clear way to reach the accuracy needed using the existing polarimeters.

- Compton: uncomfortably low energy
- Moller with Fe target: low current

Improvements Considered

- Compton: green laser (still marginal performance)
- Moller: 100% polarized atomic hydrogen target

Møller Polarimetry with Atomic Hydrogen Targets

Advantages:

- 100% electron polarization
- hydrogen target: no Levchuk effect, low background
- high beam current

Goal:

$\sim 0.5\%$ systematic error

Atomic hydrogen ground state: $\vec{\mu}_H \sim \vec{\mu}_e$

Stored in a trap at $\sim 300\text{ mK}$:

- Z: solenoid field $B_{max} \sim 5 - 8\text{ T}$, energy $\sim \vec{\mu} \cdot \vec{B}$
- R: copper cylinder $d \sim 3\text{ cm}$, superfluid He film
- Density $\sim 3 \cdot 10^{15}\text{ H/cm}^3$
- Polarization $\sim 1 - \mathcal{P} \sim \exp(\frac{-2\mu B}{kT}) \sim 10^{-15}$,
hyperfine interaction: $\sim 1 - \mathcal{P} \sim 10^{-6}$
- Lifetime: recombination (3-body coll., surfaces)
 $> 1000\text{ sec}$
- Self-cleaning time $\sim 2\text{ msec}$
- Filling the cell: several minutes

Statistical accuracy 1% in $\sim 30\text{ min}$ at $30\text{ }\mu\text{A}$

Conclusion

- Various polarimetry techniques are used in JLab
- Polarimetry errors become the dominant one for the planned parity violation experiments
- An accuracy of $\sim 1.5\%$ is achieved at 2-6 GeV
- New methods are considered for improving the accuracy in the full range of CEBAF 0.8-6 GeV. Using of atomic hydrogen may provide a superb error of $\sim 0.5\%$